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Field Case Study Report MAKSA App Validation on Frozen-Thawed Bovine Semen (Fujairah, UAE, 2025)

Date: 14 November 2025
Location: Bulaida Farms, Fujairah

Experts:

Dr. Andrés Senetiner
20+ years Cattle Specialist,
Bovine Genetics & Cloning

Maksim Mershiev
CEO&Co-Founder
of Avare Biotech

Initial AAS Signed by: Andres Alberto Senetiner 1F70234FA1064FF...
Signature

Initial MM Signed by: Maksim Mershiev 325996BF0B2B43A...
Signature

Executive Summary / Conclusions

- "Avare Biotech" app powered by MAKSA technology showed that mobile semen analysis can be accurate, consistent and practical in real farm conditions.
- Motility and progressive motility matched the CASA system, even though we worked with ordinary tools — an iPhone 14 Pro and a simple $300 microscope.
- The difference in concentration was expected. At the time of testing we had no equine or bovine frozen-thawed video data from this **specific microscope-camera** pair in the training set. Now, after adding these datasets, the concentration model continues improving.

The main outcome: the technology already works well for motility, and the remaining accuracy gap for concentration will be closed as more field videos enter the training pipeline.

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Objective

Validate how accurately the MAKSA app measures motility and concentration on frozen-thawed bovine semen when compared to a commercial CASA system (Androscope).

How the Test Was Performed

1. Dr Andrés thawed the frozen bovine semen using his usual farm protocol.
2. He checked the sample on his available Androscope CASA system and saved the CASA slide.
3. Maksim reviewed the same sample using a basic 300-dollar light microscope and an iPhone 14 Pro.
4. Video was recorded directly through the eyepiece without adaptors, filters or enhancements.
5. The video was uploaded to MAKSA and processed automatically.
6. Results from both systems were compared for concentration, total motility and progressive motility.

This setup intentionally mimicked real field conditions — simple equipment, natural light, normal farm workflow.

Sample Details

- Type: frozen-thawed bovine semen
- Dilution: undiluted
- Methods compared: MAKSA vs CASA (Androscope)

Table 1MAKSA results comparison

Tests done	Key Semen Parameters	MAKSA	CASA (Androscope)	Difference, %
Bull 6	Concentration M/ml	40	58.9	47%
	Total Motility %	55	70	27
	Progressive Motility %	41	58	39

Interpretation

- **Motility matches CASA extremely well.** This confirms that MAKSA's motion-tracking engine is ready for routine use on cattle farms, breeding centers and AI technicians.

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- **Concentration gap expected.** No prior frozen-thawed bovine dataset existed for this specific microscope and camera combination. After adding these videos, the model's accuracy will rise quickly.
- **User-friendly workflow.** The test reaffirmed that even with simple equipment, farmers can get reliable motility readings without CASA hardware.

Recommendations for Avare

1. **Recording Assistant in the App**
   A simple guided flow that helps users capture stable, clear videos.
2. **Automatic Quality Filter**
   Frame quality scoring to avoid blurry or low-light videos before analysis.
3. **Short Onboarding Module**
   A small video + checklist in the app explaining:
   - Use of 200× magnification
   - Start recording quickly after thawing
   - Avoid drying of the sample
   - Keep temperature consistent (around 36–38°C if possible)

Next Steps

- Conduct more field tests with properly thawed samples and standardized media.
- Expand testing to at least 50 additional bovine samples with varied microscopes and phones.
- Compare MAKSA results with both hardware CASA and manual counting for full validation.
- Add all new videos into the model training set to improve concentration accuracy.

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Appendix - Screenshots of CASA results and MAKSA reports

Bull 6: CASA readings

Motility & Concentration

Sample analysis

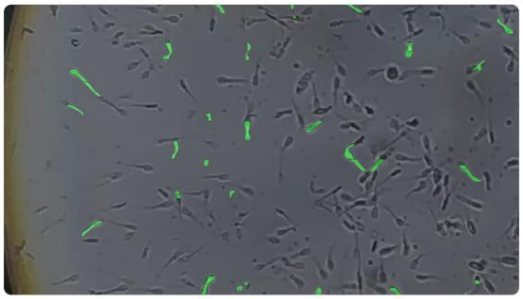
Date of analysis:		11/14/2025
Total number of sperm analyzed:		1494
Number of fields:		3
Dilution ratio:		1+0
Concentration:	[10^6/ml]	58.89
Total motility:	[%]	70.10
Progressive motility:	[%]	58.47
Fast motility:	[%]	21.50
Slow motility:	[%]	35.61
Circle motility:	[%]	1.36
Local motility:	[%]	11.63
Immotile:	[%]	29.90

Average kinematic		VCL [µm/s]	VSL [µm/s]	VAP [µm/s]	DCL [µm]	DSL [µm]	DAP [µm]	ALH [µm]	BCF [Hz]	HAC [rad]	LIN [VSL/VCL]	STR [VSL/VAP]
Total motility	[=]	92.08	42.18	50.32	28.91	13.39	16.01	2.16	12.05	0.28	0.52	0.84
	[σ]	72.81	33.40	36.82	22.54	10.53	11.69	1.85	7.95	0.25	0.32	0.28
Progressive motility	[=]	103.50	47.32	56.44	32.19	14.90	17.80	2.41	12.95	0.31	0.52	0.84
	[σ]	77.86	35.51	39.08	24.01	11.12	12.34	1.99	8.25	0.27	0.33	0.30
Fast motility	[=]	153.20	59.42	74.87	43.11	16.28	20.91	3.71	12.68	0.43	0.40	0.78
	[σ]	55.39	31.45	29.84	22.87	9.96	11.17	1.43	6.80	0.19	0.18	0.21
Slow motility	[=]	60.86	36.15	39.74	22.60	13.43	14.77	1.31	13.04	0.20	0.62	0.89
	[σ]	15.45	15.26	13.71	8.75	7.10	6.81	0.55	5.79	0.15	0.24	0.16
Circle motility	[=]	201.85	88.72	112.33	58.82	23.83	31.99	4.85	15.73	0.52	0.45	0.78
	[σ]	235.63	155.36	139.63	74.01	20.92	29.21	6.45	21.83	0.61	0.58	0.72
Local motility	[=]	30.82	14.60	17.52	11.34	5.29	6.41	0.80	7.20	0.14	0.52	0.83
	[σ]	9.71	5.83	5.57	5.40	2.94	3.16	0.31	4.94	0.11	0.23	0.18

Bull 6: MAKSA values

Analysis 14.11.2025

Animal: — **Name:** Andre's 1good **Age:** 11 **Breed:** good1 **Status:** Ready **Created:** 2025-11-14 07:37



Total objects:	25
Concentration (millions/ml^3):	40
Dilution rate:	[1:0]
Volume of field of view (mm^3):	0.001744
Progressive Motility:	41%
Non-Progressive Motility:	14%
Immotility:	45%

Agglutination

Found:	No